PRESCRIBING INFORMATION

ERANFU

Fulvestrant Solution for Injection 250 mg/5 mL

QUALITATIVE AND QUANTITATIVE COMPOSITION

Each 5 mL of prefilled syringe contains:

Fulvestrant	250 mg
Benzyl alcohol	10%w/v
Alcohol (Ethanol 96%)	10%w/v
Benzyl Benzoate	15%w/v
Castor oil	q.s.
Nitrogen	q.s.

PHARMACEUTICAL FORM

Solution for Injection

Fulvestrant Injection is a clear, colorless to yellow, viscous liquid free from visible particulate matter, presented in 5.0 mL Type-I glass syringe with OVS tip cap.

CLINICAL PARTICULARS

Therapeutic indications

Fulvestrant Injection is indicated:

• as monotherapy for the treatment of estrogen receptor positive, locally advanced or metastatic breast cancer in postmenopausal women:

- who are human epidermal growth factor receptor 2 (HER2)-negative and not previously treated with endocrine therapy.
- with disease relapse on or after adjuvant endocrine therapy, or disease progression on endocrine therapy.

• in combination with ribociclib for the treatment of hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative advanced or metastatic breast cancer in postmenopausal women, as initial endocrine based therapy or following disease progression on endocrine therapy. *

• in combination with abemaciclib for the treatment of women with hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative advanced or metastatic breast cancer with disease progression following endocrine therapy. *

• in combination with palbociclib for the treatment of hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative locally advanced or metastatic breast cancer in women who have received prior endocrine therapy (see Pharmacodynamic Properties). *

In pre- or perimenopausal women, the combination treatment with palbociclib, ribociclib or abemaciclib should be combined with a luteinizing hormone releasing hormone (LHRH) agonist.

** This indication is only applicable in markets where palbociclib, ribociclib, or abemaciclib are registered.*

Posology and method of administration

Posology

Adult females (including the elderly).

The recommended dose is 500 mg at intervals of one month, with an additional 500 mg dose given two weeks after the initial dose.

When Fulvestrant Injection is used in combination with palbociclib, abemaciclib or ribociclib, please also refer to the Summary of Product Characteristics of palbociclib, abemaciclib or ribociclib.

Prior to the start of treatment with the combination of Fulvestrant Injection plus palbociclib, abemaciclib or ribociclib, and throughout its duration, pre/perimenopausal women should be treated with LHRH agonists according to local clinical practice.

Special populations

Paediatric patient:

The safety and efficacy of Fulvestrant Injection in children from birth to 18 years of age have not been established. Currently available data are described in sections Pharmacodynamic Properties and Pharmacokinetic Properties, but no recommendation on a posology can be made.

Renal impairment:

No dose adjustments are recommended for patients with mild to moderate renal impairment (creatinine clearance ≥ 30 ml/min). Safety and efficacy have not been evaluated in patients with severe renal impairment (creatinine clearance < 30 ml/min) and, therefore, caution is recommended in these patients (see Special Warnings and Precautions for Use).

Hepatic impairment:

No dose adjustments are recommended for patients with mild to moderate hepatic impairment. However, as fulvestrant exposure may be increased, Fulvestrant Injection should be used with caution in these patients. There are no data in patients with severe hepatic impairment (see Contraindications, Special Warnings and Precautions for Use and Pharmacokinetic Properties).

Method of administration

Fulvestrant Injection should be administered as two consecutive 5 ml injections by slow intramuscular injection (1-2 minutes/injection), one in each buttock (gluteal area).

Caution should be taken if injecting Fulvestrant Injection at the dorsogluteal site due to the proximity of the underlying sciatic nerve.

For detailed instructions for administration, see Instructions for administration and Special precautions for disposal.

Instructions for administration

Administer the injection according to the local guidelines for performing large volume intramuscular injections.
NOTE: Due to the proximity of the underlying sciatic nerve, caution should be taken if administering Fulvestrant at the dorsogluteal injection site.

Warning - Do not autoclave safety needle (SurGuard®3 Safety Hypodermic Needle) before use. Hands must remain behind the needle at all times during use and disposal.

For each of the two syringes:

- Remove glass syringe barrel from blister tray and check that it is not damaged (see figure 1)

Figure 1

- Break the seal of the white plastic cover on the syringe Luer connector Luer-Lok to remove the cover with the attached rubber tip cap (see Figure 2)

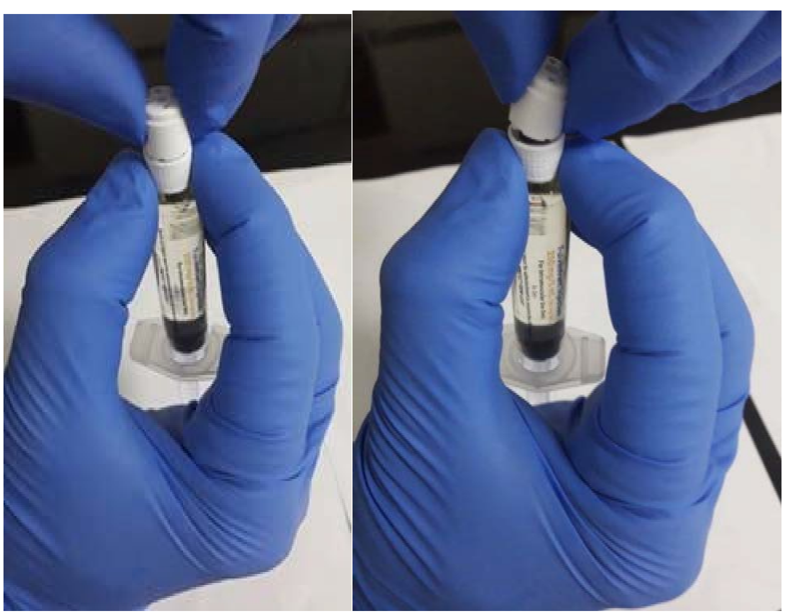

Figure 2

- Peel open the safety needle (Terumo® SurGuard) outer packaging (see Figure 3)

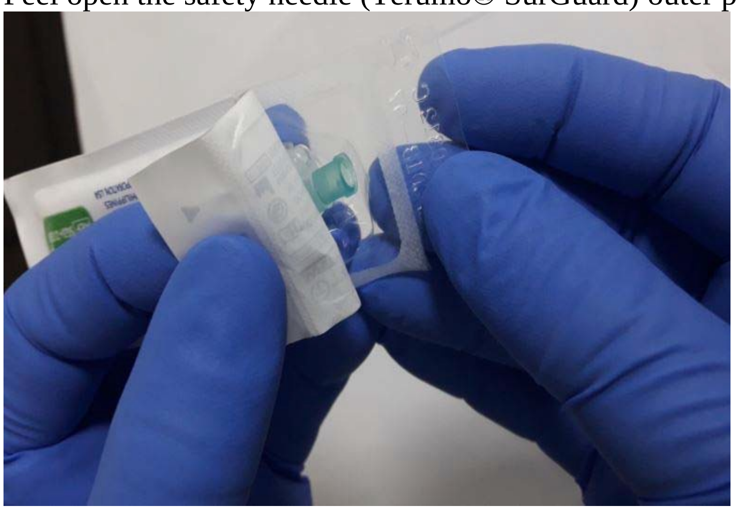

Figure 3

- Attach the safety needle to the Luer-Lok (see figure 4)

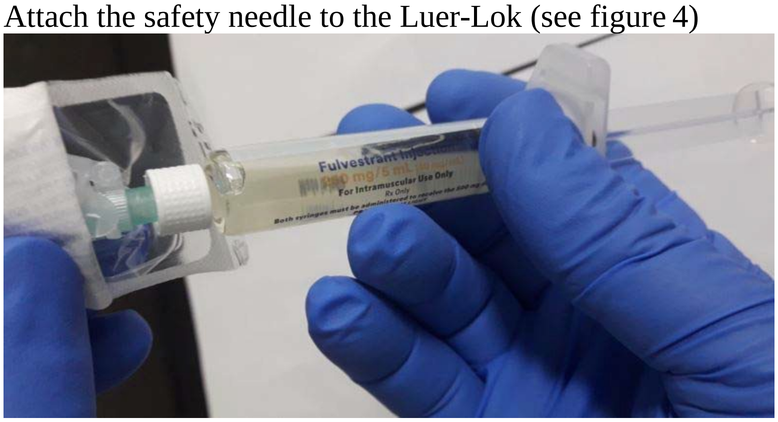

Figure 4

- Twist until firmly seated.
- Twist to lock the needle to the Luer connector.
- Transport filled syringe to point of administration.

- Move the safety sheath away from the needle and toward the syringe barrel to the angle shown, prior to removing the needle cap (see figure 5)

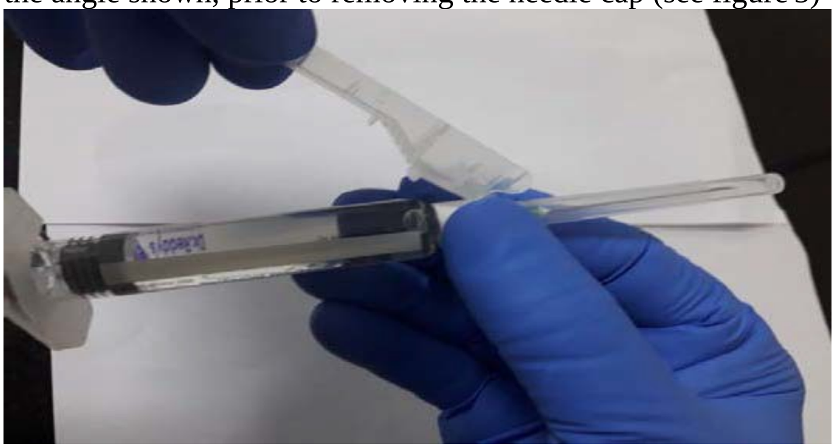

Figure 5

- Parenteral solutions must be inspected visually for particulate matter and discolouration prior to administration.
- Remove the needle cap (see figure 6)

Figure 6

- Expel excess gas from the syringe.
- Administer intramuscularly slowly (1-2 minutes/injection) into the buttock. For user convenience, the needle bevel- up position is oriented to the lever arm (see figure 7)

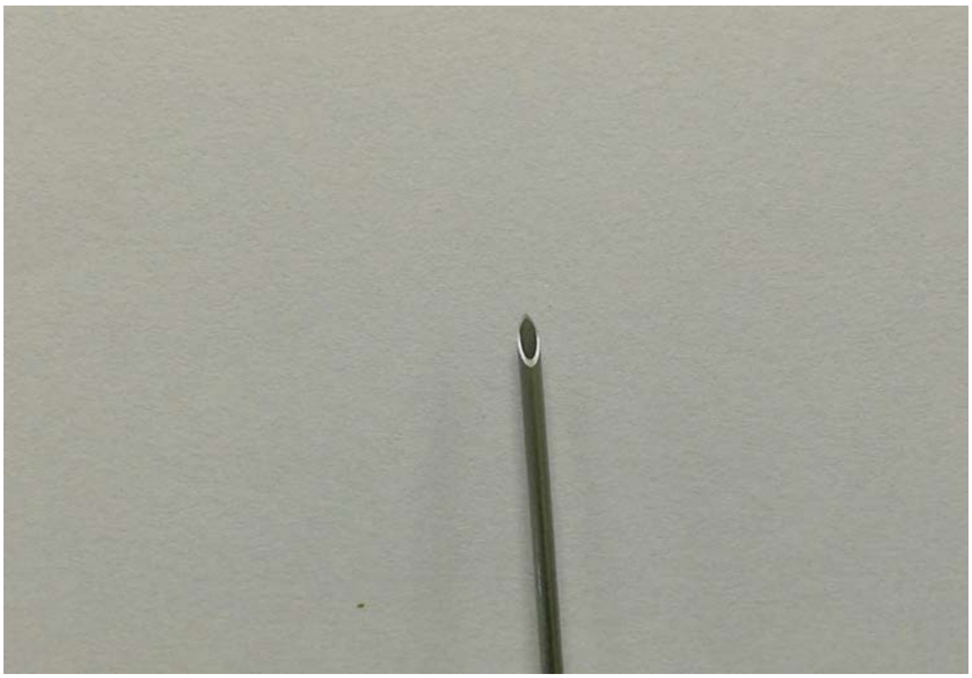

Figure 7

- Use the finger grip if necessary.
- After injection, use a one-handed technique to activate the safety mechanism using any of the two methods illustrated below (see figure 8 and 9) (Activation is verified by an audible and/or tactile “click” and can be visually confirmed.) NOTE: Activate away from self and others. Listen for click and visually confirm needle tip is fully covered (see figure 10).

Finger Activation:

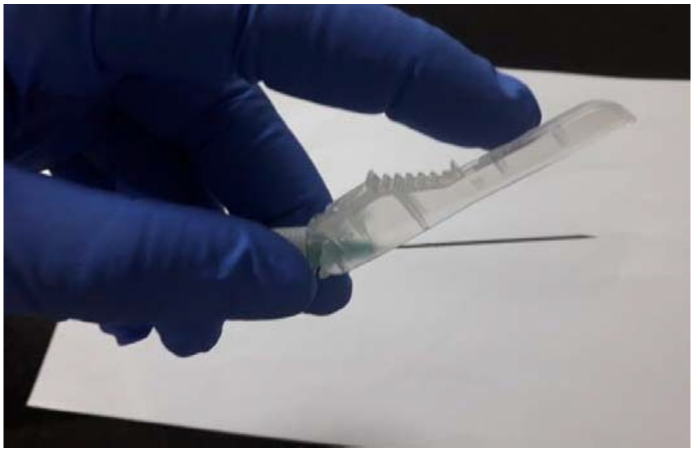

Figure 8

Thumb Activation:

Figure 9

Visual Confirmation:

Figure 10

Disposal:

Pre-filled syringes are for single use **only**.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

Contraindications

- Hypersensitivity to the active substance, or to any of the other excipients.
- Pregnancy and lactation.
- Severe hepatic impairment.

Special Warnings and Precautions for Use

As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age. Not to be used in neonates.

Fulvestrant should be used with caution in patients with mild to moderate hepatic impairment.

Fulvestrant should be used with caution in patients with severe renal impairment (creatinine clearance less than 30 ml/min).

Due to the intramuscular route of administration, fulvestrant should be used with caution if treating patients with bleeding diatheses, thrombocytopenia or those taking anticoagulant treatment.

Thromboembolic events are commonly observed in women with advanced breast cancer. This should be taken into consideration when prescribing fulvestrant to patients at risk.

Injection site related events including sciatica, neuralgia, neuropathic pain and peripheral neuropathy have been reported with fulvestrant injection. Caution should be taken while administering fulvestrant at the dorsogluteal injection site due to the proximity of the underlying sciatic nerve.

There are no long-term data on the effect of fulvestrant on bone. Due to the mechanism of action of fulvestrant, there is a potential risk of osteoporosis.

Interference with estradiol antibody assays

Due to the structural similarity of fulvestrant and estradiol, fulvestrant may interfere with antibody based-estradiol assays and may result in falsely increased levels of estradiol.

Paediatric population

Fulvestrant is not recommended for use in children and adolescents as safety and efficacy have not been established in this group of patients.

Interaction with other medicinal products and other forms of interaction

A clinical interaction study with midazolam (substrate of CYP3A4) demonstrated that fulvestrant does not inhibit CYP3A4. Clinical interaction studies with rifampicin (inducer of CYP3A4) and ketoconazole (inhibitor of CYP3A4) showed no clinically relevant change in fulvestrant clearance. Dose adjustment is therefore not necessary in patients who are receiving fulvestrant and CYP3A4 inhibitors or inducers concomitantly.

Fertility, pregnancy and lactation

Women of child bearing potential

Patients of child-bearing potential should be advised to use effective contraception while on treatment.

Pregnancy

Fulvestrant is contraindicated in pregnancy. If pregnancy occurs while taking fulvestrant, the patient must be informed of the potential hazard to the foetus and potential risk for loss of pregnancy.

Breast-feeding

Breast-feeding must be discontinued during treatment with fulvestrant. It is not known whether fulvestrant is excreted in human milk. Considering the potential for serious adverse reactions due to fulvestrant in breast-fed infants, use during lactation is contraindicated.

Fertility

The effects of fulvestrant on fertility in humans has not been studied.

Effects on ability to drive and use machines

Fulvestrant has no or negligible influence on the ability to drive or use machines. However, since asthenia has been reported very commonly with fulvestrant, caution should be observed by those patients who experience this adverse reaction when driving or operating machinery.

Undesirable Effects

The most frequently reported adverse reactions are injection site reactions, asthenia, nausea, and increased hepatic enzymes (ALT, AST, ALP).

Table: Adverse Drug Reactions

Adverse reactions by system organ class and frequency		
Infections and infestations	Common	Urinary tract infections
Blood and lymphatic system disorders	Common	Reduced platelet count
Immune system disorders	Very common	Hypersensitivity reactions
	UnCommon	Anaphylactic reactions
Metabolism and nutrition disorders	Common	Anorexia
Nervous system disorders	Common	Headache
Vascular disorders	Very common	Hot flushes
	Common	Venous thromboembolism
Gastrointestinal disorders	Very common	Nausea
	Common	Vomiting, diarrhoea
Hepatobiliary disorders	Very common	Elevated hepatic enzymes (ALT, AST, ALP)
	Common	Elevated bilirubin
	Uncommon	Hepatic failure, hepatitis, elevated gamma-GT
Skin and subcutaneous tissue disorders	Very Common	Rash

Musculoskeletal and connective tissue disorders	Very Common	Joint and musculoskeletal pain (Includes: arthralgia, and less frequently musculoskeletal pain, myalgia and pain in extremity.)
	Common	Back pain
Reproductive system and breast disorders	Common	Vaginal haemorrhage
	Uncommon	Vaginal moniliasis, leukorrhoea,
General disorders and administration site conditions	Very common	Asthenia, injection site reactions (The term injection site reactions does not include the terms injection site haemorrhage, injection site haematoma, sciatica, neuralgia and neuropathy peripheral.)
	Common	sciatica, peripheral neuropathy
	Uncommon	Injection site haemorrhage, injection site haematoma, neuralgia

Overdose

There are isolated reports of overdose with fulvestrant in humans. Animal studies suggest that no effects other than those related directly or indirectly to anti-oestrogenic activity were evident with higher doses of fulvestrant. If overdose occurs, symptomatic supportive treatment is recommended.

PHARMACOLOGICAL INFORMATION

Pharmacodynamic properties

Mechanism of action and pharmacodynamic effects

Fulvestrant is a competitive oestrogen receptor (ER) antagonist with an affinity comparable to oestradiol. Fulvestrant blocks the trophic actions of oestrogens without any partial agonist (oestrogen-like) activity. The mechanism of action is associated with down-regulation of oestrogen receptor protein levels.

Pharmacokinetic properties

Absorption

After administration of fulvestrant long-acting intramuscular injection, fulvestrant is slowly absorbed and maximum plasma concentrations (C) are reached after about 5 days. max

Distribution

Fulvestrant is subject to extensive and rapid distribution. The large apparent volume of distribution at steady state (Vd) of approximately 3 to 5 l/kg suggests that distribution is largely extravascular. Fulvestrant is is highly (99%) bound to plasma proteins. Very low density lipoprotein (VLDL), low density lipoprotein (LDL), and high density lipoprotein (HDL) fractions are the major binding components. No interaction studies were conducted on competitive protein binding. The role of sex hormone-binding globulin (SHBG) has not been determined.

Biotransformation

The metabolism of fulvestrant has not been fully evaluated, but involves combinations of a number of possible biotransformation pathways analogous to those of endogenous steroids. Identified metabolites (includes 17-ketone, sulphone, 3-sulphate, 3- and 17-glucuronide metabolites) are either less active or exhibit similar activity to fulvestrant in anti-oestrogen models.

Elimination

Fulvestrant is eliminated mainly in metabolised form. The major route of excretion is via the faeces, with less than 1% being excreted in the urine. Fulvestrant has a high clearance, 11±1.7 ml/min/kg, suggesting a high hepatic extraction ratio. The terminal half-life (t) after intramuscular administration is governed by the absorption rate and was estimated to be 50 days.

Incompatibilities

In the absence of incompatibility studies, this medicinal product must not be mixed with other medicinal products.

PHARMACEUTICAL INFORMATION

Packaging available (pack size)

Each pack size contains:

- 2 Prefilled syringes, each containing 250 mg/5 mL of Fulvestrant solution,

- 2 Safety Hypodermic 21G, 1½ inch Needles with Plastic needle Shield, placed in a tray in a carton.

Shelf Life: 24 Months

Please refer to expiry date on label/outer carton.

Keep medicine out of reach of children

Precautions for Storage:

Store in refrigerator at 2°C to 8°C, Do not freeze

Store the pre-filled syringe in the original package, in order to protect from light.

Nature and Contents of container

2 pre-filled syringes in a carton

Manufactured by

Dr. Reddy’s Laboratories Limited
FTO-IX, Plot No. Q1 to Q5, Phase-III, VSEZ,

Duvvada, Visakhapatanam-530 046,
Andhra Pradesh, India**.**

Date of revision

August 2021